

# SCARLETT ANDERSON

Principal Software Engineer | AI-Driven Systems | Full Stack Development

✉ Email 🔗 linkedin.com 📍 San Jose, California

## SUMMARY

Principal Software Engineer with 9+ years of experience building and deploying scalable, production-grade systems. Skilled in full-stack development (React, TypeScript, Python, FastAPI) and cloud infrastructure (AWS, Kubernetes, Terraform), with hands-on exposure to AI frameworks and scientific computing. Known for driving cross-functional collaboration with ML researchers, platform engineers, and scientists to deliver secure, high-performance applications that accelerate AI-driven research.

## CORE COMPETENCIES

**Full Stack Development & Architecture • Cloud & DevOps Engineering •**

**AI & Scientific Computing • System Design & Scalability •**

**Observability & Reliability • Technical Leadership •**

**Orchestration & Workflow Automation • Security & Compliance**

## EXPERIENCE

### Principal Software Engineer

Google 06/2020 - Present Mountain View, California

- Spearheaded the complete software development lifecycle for an AI-driven scientific platform, implementing architecture that scaled effortlessly and met an uptick in demand by 80%.
- Designed and developed robust full-stack web applications using modern technologies like React, Python, and AWS, enhancing scientists' abilities to analyze data effectively.
- Collaborated with ML researchers and domain scientists to integrate advanced machine learning models into our software, significantly increasing research outcomes.
- Established best practices for code quality and documentation, which improved overall team performance and reduced debugging time by 45%.
- Operationalized ML models in a production environment by using observability tools to gauge performance and continuously enhance our applications.

### Senior Software Engineer

IBM 05/2016 - 05/2020 San Jose, California

- Developed and maintained scalable cloud infrastructure solutions utilizing AWS and Kubernetes, ensuring high availability and efficiency across all services.
- Led a cross-functional team of software developers and product managers to create advanced data processing applications that improved data insights by 62%.
- Implemented containerization using Docker and Terraform, leading to streamlined deployment processes and a 40% reduction in operational costs.
- Conducted workshops for training junior developers on best industry practices, resulting in improved team skill level and project delivery timelines.

### Software Engineer

Microsoft 08/2013 - 04/2016 Redmond, Washington

- Created web applications using full-stack technologies that supported various teams in the implementation of ML-driven solutions, enabling better decision making.
- Worked closely with scientists and engineers to develop users' requirements and convert them into functional applications, which increased user satisfaction by 65%.
- Participated in agile development processes, significantly contributing to a more adaptive software development environment and improving project turnaround times.
- Led the introduction of RESTful APIs into legacy applications, enabling modern integrations that increased third-party collaboration and data exchange.

## KEY ACHIEVEMENTS

### Improved Development Efficiency

Led a team to redesign the CI/CD pipeline, resulting in a 50% reduction in deployment time and improved code quality.

### Scalable Application Architecture

Architected a new AI-driven platform that scaled to support over 100,000 queries per day, significantly enhancing user experience.

### Cross-Functional Collaboration

Successfully initiated and executed a collaboration project between software engineers and data scientists, resulting in a 30% increase in innovative solutions deployed.

### Mentorship in Technical Leadership

Mentored junior engineers leading to a 35% increase in team productivity by improving their coding skills and best practices understanding.

## EDUCATION

### Master of Science in Computer Science

Stanford University  
2013 Stanford, California

## ADDITIONAL TRAINING

### Deep Learning Specialization

A five-course specialization from Coursera that provides a comprehensive understanding of deep learning techniques and their applications.

### Cloud Computing for Business

Earned this certification from edX focusing on cloud services and deployment strategies for scaling business applications.